Healthy Skepticism Session 4

If two people have the same diagnosis, what do they actually have in common?

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About Us

Dr. Murray Cantor

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- Not anti-vax or MAHA
- From that perspective, there is a lot of confusion about medical claims
- Respect for the challenge of medical research

- Practicing physician with direct patient-care experience
- Translates clinical evidence into plain-language guidance
- Shares a commitment to honest, balanced medical communication

Disclaimer

This presentation is for educational purposes only.

The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.

The Six Faults

1	**Relative reporting**	*What does a 36% improvement from taking a drug really mean?*
2	**Probability reversal**	*Does a positive COVID test mean I have it? Does a negative mean I'm not contagious?*
3	**Arbitrary categories**	*How concerned should you be if your systolic pressure goes from 119 to 120?*
4	**The Bernoulli fallacy**	*Why does the published medical advice keep changing?*
5	**The causal leap**	*Does bird watching prevent dementia?*
6	**The syndrome trap**	*If two people have the same diagnosis, what do they actually have in common?*

Today: the syndrome trap. Final session: the causal leap.

FAULT 6

The Syndrome Trap

Confusing a Syndrome, or a Spectrum, with a Disease

A Disease Is One Thing. A Syndrome Is a Checklist.

DISEASE

One mechanism.

Because there is a mechanism, there is a reasonably predictable course, and a treatment aimed at the cause.

Example: strep throat. One bacterium, one test, one targeted antibiotic.

SYNDROME

One name for a cluster.

A committee assigns the name, criteria decide the membership, and the criteria are checked off a list.

Naming a cluster is useful. It is how medicine starts to study something it does not yet understand.

Some syndromes later graduate to diseases, once a mechanism is found. The trap is talking as if that already happened.

The fault: treating a checklist as if it were a mechanism, with one prognosis and one treatment.

Illustrative framing; the three cases that follow are each fully sourced.

Metabolic Syndrome: The Checklist

You have metabolic syndrome if you meet any three of these five criteria.

Criterion	Line that puts you over
Waist circumference	40 in or more for men, 35 or more for women
Triglycerides	150 mg/dL or more, or on treatment
HDL cholesterol	Under 40 mg/dL for men, under 50 for women, or on treatment
Blood pressure	130/85 mm Hg or more, or on treatment
Fasting glucose	100 mg/dL or more, or on treatment

Each line is a threshold drawn on a continuous measurement. Fault 3 is at work five times before the syndrome is even assembled.

Five committee-drawn lines, and a rule: any three of five.

Grundy SM et al. Diagnosis and management of the metabolic syndrome: AHA/NHLBI scientific statement. Circulation 2005;112:2735–2752.

Metabolic Syndrome: Counting the Ways In

Any three of five qualifies. Each row is a distinct patient type; here is every one, in order, so you can check the count.

	Waist	Trig	HDL	BP	Glucose
1	●	●	●	○	○
2	●	●	○	●	○
3	●	●	○	○	●
4	●	○	●	●	○
5	●	○	●	○	●
6	●	○	○	●	●
7	○	●	●	●	○
8	○	●	●	○	●
9	○	●	○	●	●
10	○	○	●	●	●

Filled dot = over the line for that criterion.

The syndrome identifies ten different patient types, and gives them all one name.

The any-three-of-five rule from the AHA/NHLBI criteria (Grundy et al. 2005). The ten rows are exhaustive and in systematic order.

Metabolic Syndrome: A Treatment Plan for Each Type

Read each row's filled dots as that patient's chart. What the chart calls for changes from row to row.

	Waist	Trig	HDL	BP	Glucose	What the chart calls for
1	●	●	●	○	○	Weight · Lipids
2	●	●	○	●	○	Weight · Lipids · Pressure
3	●	●	○	○	●	Weight · Lipids · Glucose
4	●	○	●	●	○	Weight · Lipids · Pressure
5	●	○	●	○	●	Weight · Lipids · Glucose
6	●	○	○	●	●	Weight · Pressure · Glucose
7	○	●	●	●	○	Lipids · Pressure
8	○	●	●	○	●	Lipids · Glucose
9	○	●	○	●	●	Lipids · Pressure · Glucose
10	○	○	●	●	●	Lipids · Pressure · Glucose

Each patient type calls for its own treatment plan. No single prescription fits the label.

Treatment mapping follows the AHA/NHLBI direction to treat the components (Grundy et al. 2005): waist → weight management; triglycerides and HDL → lipid therapy; blood pressure → antihypertensives; glucose → glucose control.

Metabolic Syndrome: There Is No Pill for the Label

Nearly four decades after the syndrome was named, no drug carries it as an approved indication.

AHA / NHLBI, 2005

Treat the components.

The scientific statement that defines the syndrome directs clinicians to treat the pressure, the lipids, the glucose, and the weight, each on its own evidence.

ADA / EASD, 2005

"Time for a critical appraisal."

The diabetes establishment, reviewing its own label: the criteria are ambiguous, the label predicts little beyond its parts, and clinicians should treat the parts.

The people who defined the syndrome do not treat the syndrome.

Grundy et al., Circulation 2005;112:2735–2752. Kahn R, Buse J, Ferrannini E, Stern M. Diabetes Care 2005;28:2289–2304.

Metabolic Syndrome: What Failed and What Worked

Two ways medicine has aimed at this syndrome, with opposite results.

TREATING THE CHECKLIST

It failed.

Two drug programs raised HDL, one of the five criteria. One raised deaths (torcetrapib). The other helped nobody (niacin).

TREATING A MECHANISM

It arrived by accident.

GLP-1 drugs (Ozempic) improve every criterion. They came out of diabetes safety trials, not out of the syndrome.

The drug works because it acts upstream, on a mechanism the checklist never named.

The label did not find the mechanism. The drug did.

Barter PJ et al., NEJM 2007;357:2109–2122. AIM-HIGH, NEJM 2011;365:2255–2267. FDA guidance, Dec 2008. Marso SP et al. (LEADER), NEJM 2016;375:311–322. Lincoff AM et al. (SELECT), NEJM 2023;389:2221–2232.

Autism: One Label Since 2013

A spectrum is a syndrome with the variation admitted.

DSM-5, 2013

Several diagnoses became one.

The fifth edition of the psychiatric diagnostic manual folded what had been separate diagnoses, including autistic disorder and Asperger's disorder, into a single autism spectrum disorder.

There were defensible reasons: the old boundaries between the diagnoses were not reliably drawn. But one label now spans an enormous range of people.

Merging the labels fixed a boundary problem and created a heterogeneity problem.

American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, 5th ed., 2013.

Autism: The Same Grid, Now in Traits

The metabolic picture again: different profiles, one label. The columns are traits now, not lab lines, and there is no checklist to count.

	Language	Support	Social	Interests	Sensory
A non-verbal eight-year-old	●	●	●	○	●
A mainstreamed teenager	○	○	●	○	●
An adult diagnosed at 40	○	○	●	●	●
A chief executive	○	○	●	●	○

Filled dot = prominently affected. Rows are illustrative composites; the range itself is documented.

Whatever the first row and the last row have, it is not one disease.

The last row: Elon Musk stated his Asperger's diagnosis on Saturday Night Live, NBC, May 8, 2021, a diagnosis folded into autism spectrum disorder by DSM-5 (APA 2013).

Autism: The New Vocabulary Walks Back

The field's own language has been retreating from the single-disease picture.

THE RESEARCH

No single explanation.

A 2006 review argued the evidence supports no single explanation for autism at the genetic level or any other, and the manual itself scores severity separately on two different symptom domains.

THE CLINIC

“A multi-dimensional spectrum.”

Practitioners increasingly describe a space of traits rather than a line. More honest, and also a quiet surrender: a cluster of correlated dimensions with a committee-drawn boundary is exactly what a syndrome is.

The new vocabulary has walked back to the old concept: a syndrome.

Happé F, Ronald A, Plomin R. Nat Neurosci 2006;9:1218–1220. DSM-5 severity specifiers, APA 2013.

Dementia: The Same Grid, Now in Diseases

A third time: the columns are now the distinct diseases under one word. Alzheimer's alone is roughly 60 to 70 percent of cases.

	Alzheimer's	Vascular	Lewy body	FTD
Patient 1	●	○	○	○
Patient 2	○	●	○	○
Patient 3	○	○	●	○
Patient 4	○	○	○	●
Patient 5	●	●	○	○

Filled dot = the disease process actually present. Patients are illustrative; the mixtures are not: in community autopsy series the most common finding is mixed pathology, like Patient 5.

Five patients, one word, and the word does not say which disease anyone has.

World Health Organization, dementia fact sheet (Alzheimer's share). Schneider JA et al., Neurology 2007;69:2197–2204 (mixed pathology most common at autopsy).

Dementia: Same Label, Opposite Medicine

This is the case where treating at the label level can actively harm.

CHOLINESTERASE INHIBITORS

Help one disease under the umbrella.

Approved for Alzheimer's; no drug is approved for vascular dementia.

ANTIPSYCHOTICS

Dangerous under one wing.

In Lewy body dementia, severe neuroleptic sensitivity makes label-level prescribing hazardous.

ANTI-AMYLOID DRUGS, 2023

Mechanism, not label.

Lecanemab works only in early Alzheimer's with confirmed amyloid. Precision arriving by mechanism.

THE PATTERN

The GLP-1 shape again.

Progress comes when medicine targets a mechanism inside the label, never the label itself.

Under one label: a drug that helps, a drug that harms, and a drug that requires knowing which disease you have.

O'Brien JT, Thomas A. Lancet 2015;386:1698–1706. McKeith IG et al., Neurology 2017;89:88–100. van Dyck CH et al. (CLARITY AD), N Engl J Med 2023;388:9–21.

“A Cure for Dementia” Is a Category Error

Headlines promise progress against the umbrella. Mechanisms live under it.

WHAT THE HEADLINE HIDES

A cure for which disease?

A claim about “dementia” averages over Alzheimer's, vascular disease, Lewy body disease, frontotemporal disease, and their mixtures. A drug aimed at amyloid cannot fix a stroke. When the label is a crowd, a promise made to the label is a promise made to nobody in particular.

Hold that thought for next session: “Bird watching prevents dementia” makes a causal claim about this same crowd.

Before believing a claim about a label, ask which disease under the label it could possibly be about.

Framing follows the case sources on the two previous slides.

What To Ask When You Hear a Syndrome Name

QUESTION 1

Which components do I have?

And how far past each line am I? The label says three-of-five; your chart says which three, and by how much.

QUESTION 2

What about people like me?

What is known about people with my components, as opposed to people with my label?

QUESTION 3

What is the treatment aimed at?

A component, a mechanism, or just the name? Only the first two have evidence behind them.

Ask what the people under a label share. Sometimes a mechanism, and it's a disease. Often a checklist.

The three questions summarize the case evidence cited earlier in this session.

Today, and Next Session

- A syndrome is a checklist with a name; a disease has a mechanism
- Metabolic syndrome: ten patient types under one label, no pill for the label, and the working treatment arrived by accident
- Autism: one diagnosis spans a non-verbal child and a chief executive; “multi-dimensional spectrum” is a syndrome renamed
- Dementia: an umbrella over different diseases, where label-level treatment can harm
- The exposing question: if two people have the same diagnosis, what do they actually have in common?

Next session, the final fault: the causal leap. Does bird watching prevent dementia? Now you know what “dementia” is.

Questions

Materials for this session, including a full paper on the syndrome trap, are at healthy-skepticism.com.

Backup — ADHD: The Purest Specimen

A checklist of checklists: six or more of nine inattention symptoms, or six or more of nine hyperactivity symptoms.

THE COUNT

130 combinations per list.

And the sharper fact needs no combinatorics: a purely inattentive presentation and a purely hyperactive one can share not a single symptom. Metabolic syndrome guarantees one shared criterion. ADHD guarantees nothing.

THE TREATMENT TEST

Stimulants prove nothing.

Stimulants sharpen attention in nearly everyone: shown in healthy children at NIH in 1978. A drug that works on a trait says nothing about whether the label carves out a disease.

Two people, one diagnosis, zero shared symptoms.

DSM-5, APA 2013 (criteria). Rapoport JL et al., Science 1978;199:560–563.

Backup — Prediabetes: A Band on One Lab Value

The same trap on a single continuous measurement.

2003

The line moved, the people didn't.

The American Diabetes Association lowered the impaired-fasting-glucose line from 110 to 100 mg/dL. People crossed into the label overnight while their blood stayed exactly the same.

Some people in the band progress to diabetes and many do not, but the name argues before the evidence arrives: it sounds like the early stage of one disease.

When membership depends on where a committee draws a line, counts of the label track the line as much as the biology.

Genuth S et al., Expert Committee follow-up report. Diabetes Care 2003;26:3160–3167.